Cold-Chain Logistics AI-Assistant

Autonomous AI copilot ensuring cold-chain compliance and providing natural language analytics system

Agenda

The Business Problem

The high cost of cold-chain failures and perishable spoilage.

Operational Workflow

How dispatchers manage shipments from intake to real-time intervention.

The Solution (AI Assistant)

Autonomous multi-tool reasoning combining telemetry, weather, and SOP compliance.

Governance & Security

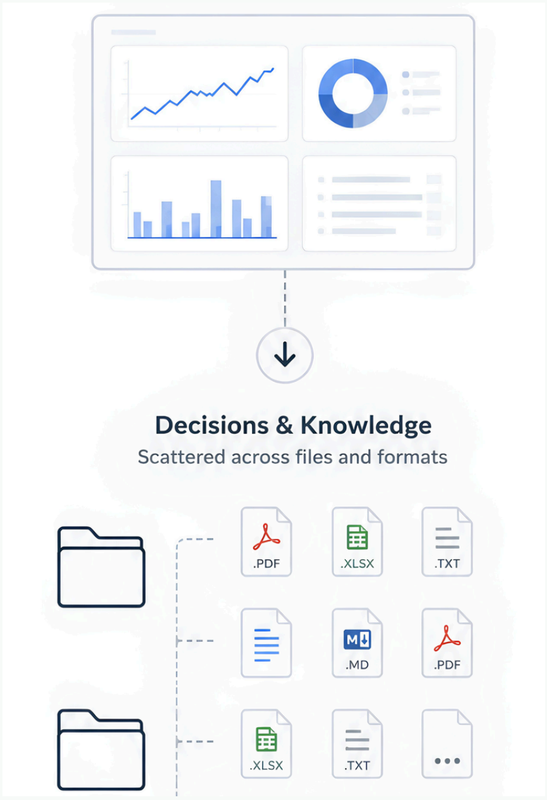
Role-based database access, human-in-the-loop validation, and immutable audit logs.

The Business Problem

The High Cost of Spoilage

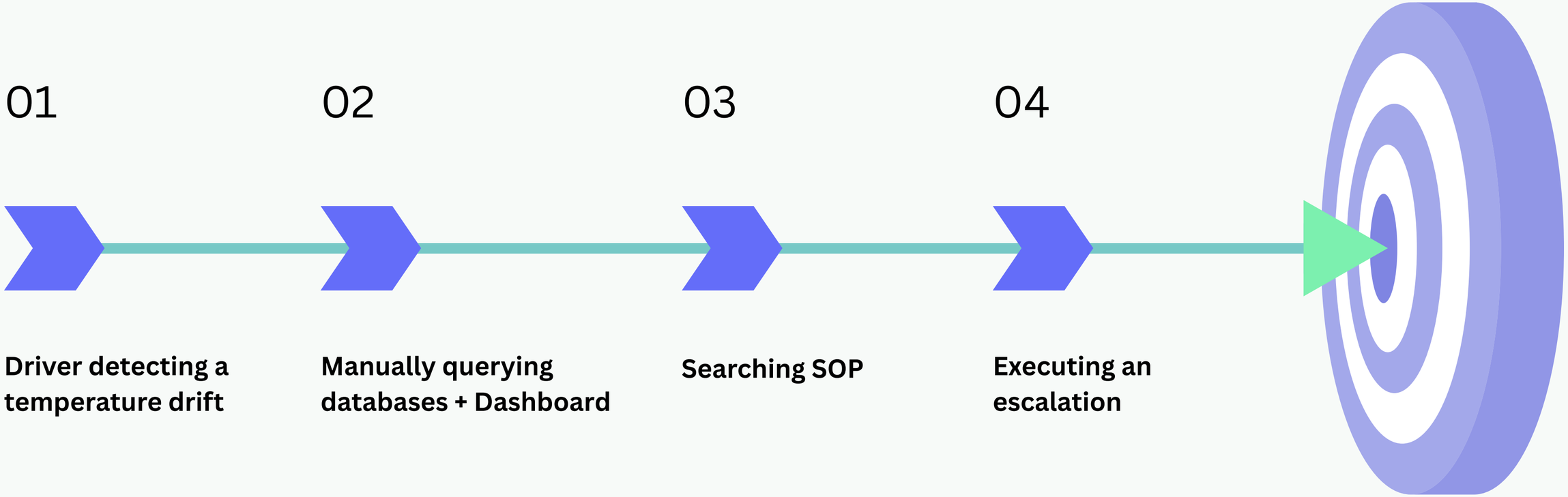


Decision from frequently updated SOP’s



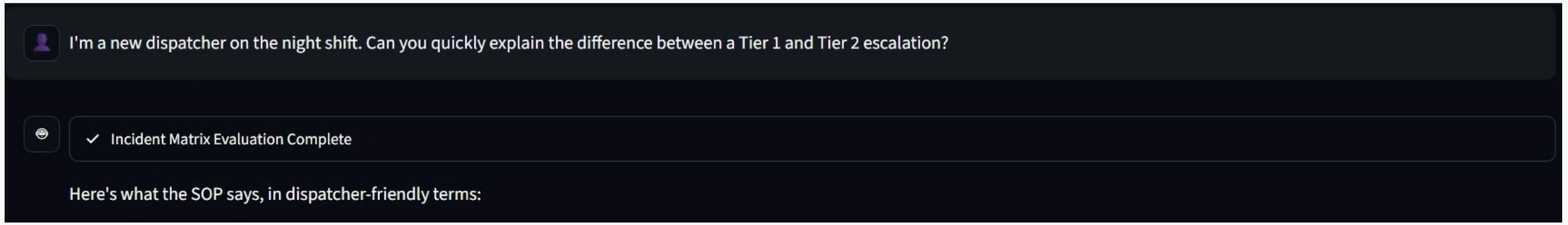
Operational Bottleneck

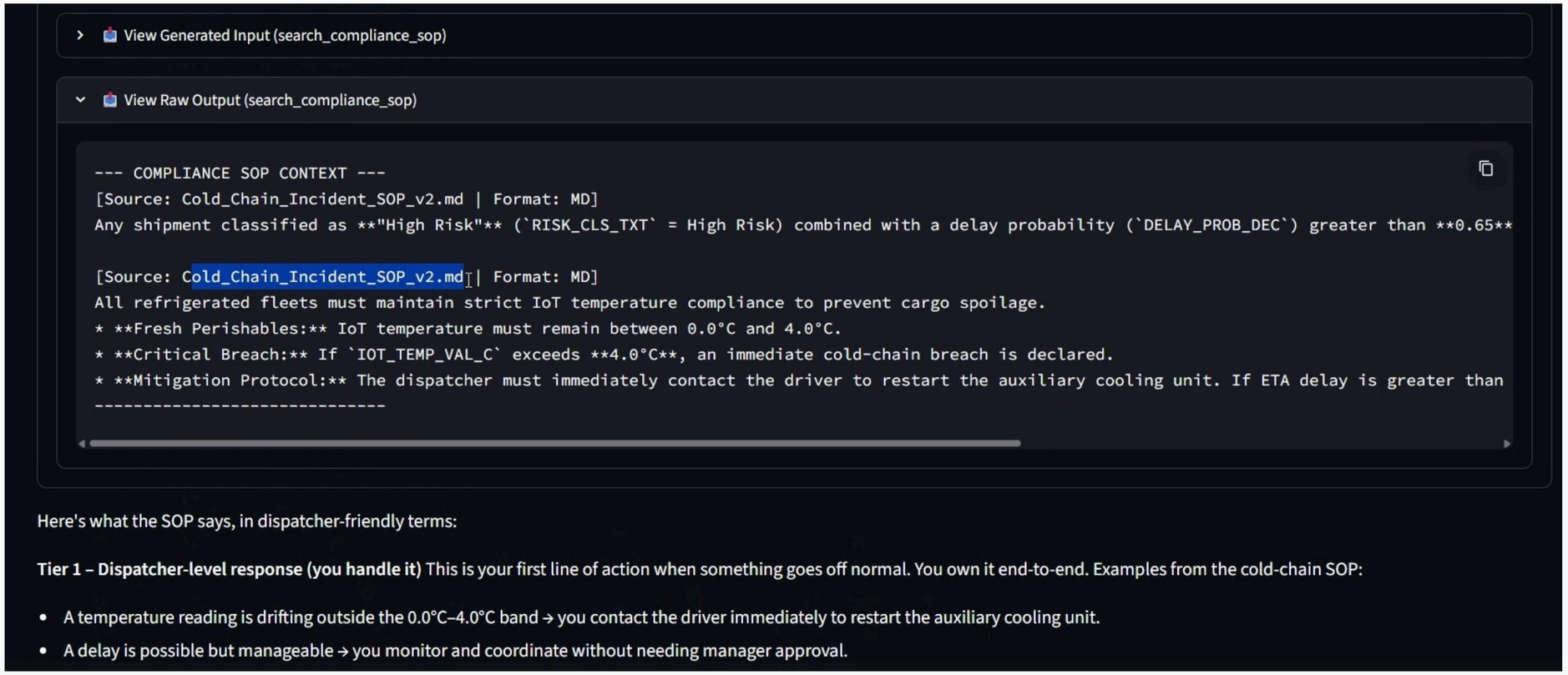
The Cost of Manual Friction



The Solution

Conversational Compliance Agent





The Solution

Conversational Compliance Agent

Tier 1 – Dispatcher-level response (you handle it) This is your first line of action when something goes off normal. You own it end-to-end. Examples from the cold-chain SOP:

- A temperature reading is drifting outside the 0.0°C–4.0°C band → you contact the driver immediately to restart the auxiliary cooling unit.
- A delay is possible but manageable → you monitor and coordinate without needing manager approval.

You are expected to resolve these yourself and document the mitigation.

Tier 2 – Escalation to the Logistics Manager (you hand it up) The SOP is very explicit about the hard trigger here:

> Any shipment classified as **"High Risk"** combined with a delay probability **greater than 0.65** must be escalated to the **Tier 2 Logistics Manager**.

So the automatic conditions for Tier 2 are:

1. `Risk_Classification` = **High Risk, AND**
2. `Delay_Probability` **> 0.65**

Also note: even without that combo, a *critical* cold-chain breach (>4.0°C) with an ETA delay of **more than 1 hour** means you must divert the vehicle to the nearest emergency cold-storage facility — and that's the kind of decision you should flag to the Tier 2 manager as well.

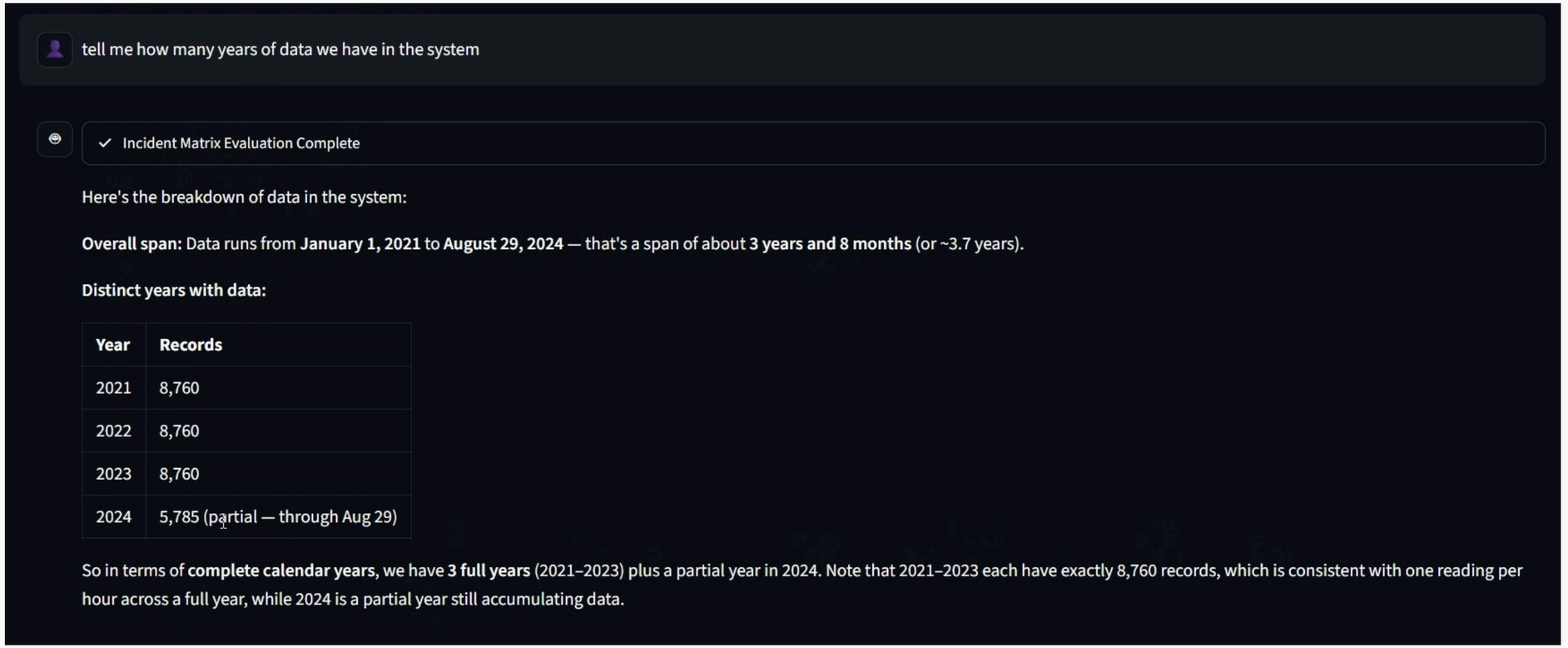
Quick rule of thumb: If you can fix it by calling the driver and rerouting on your own authority, it's Tier 1. If it's a High-Risk shipment that's likely to be delayed (>0.65 probability), or a breach with a long delay, that's Tier 2 — get the Logistics Manager involved.

Since you're on night shift, the practical tip: don't hesitate to escalate on the High-Risk + high delay probability combo — that's a hard requirement, not a judgment call.

Want me to run a quick query against the active fleet view so you can see which current shipments are sitting at or near that Tier 2 threshold?

The Solution

Conversational Compliance Agent



Governance & Security

Essential elements that build successful team collaboration



01 **Least-privilege role**

Giving AI least privileged roles

02 **Building views**

Instead of giving AI access to whole thing, the role will only have controlled visibility

03 **Immutable log table**

Every single reasoning step, tool call, and compliance check is automatically recorded to an immutable SQL audit log

04 **Role Clarity**

Defined responsibilities prevent overlap and ensure accountability for deliverables.

Thankyou